

DEPARTMENT OF MATHEMATICS & COMPUTING
Mid Semester Examination

Session: Winter Semester, 2024-2025
Subject: Engineering Mathematics-II (NMCI102)

Duration: 2 Hours
Marks: 30

Instructions:

- All questions are compulsory. **Calculators** are not allowed.
- Figures in the margin indicate marks for the concerned questions.
- Notations have their usual meaning.

1. Solve the system of linear equations

[4]

$$\begin{aligned}x + y + z - 3w &= 1 \\2x + 4y + 3z + w &= 3 \\3x + 6y + 4z - 2w &= 4.\end{aligned}$$

2. Consider the following subsets of $\mathbb{R}^4$:

- $S_1 = \{(1, -1, 2, 0), (3, 0, 0, 1), (2, 1, -1, 0)\}$
- $S_2 = \{(1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), (0, 1, 1, 1), (1, 1, 1, 1)\}$.

Examine whether S_1 and S_2 are linearly independent. Does S_1 span $\mathbb{R}^4$? Justify your answer.

[4]

3. Let $D(0, 1)$ be the set of all real-valued differentiable functions defined on interval $(0, 1)$. Let

$$V = \{y \in D(0, 1) \mid \frac{dy}{dx} + p(x)y = 0\}$$

and

$$W = \{y \in D(0, 1) \mid \frac{dy}{dx} + p(x)y = q(x)\},$$

where $p(x)$ and $q(x)$ are nonzero continuous functions on $(0, 1)$. Discuss whether V and W are vector spaces over $\mathbb{R}$ with respect to the usual addition and scalar multiplication.

[4]

4. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined as $T(x, y, z) = (2x + y + 2z, x + 2y + z)$. Is T linear? If yes, then find the bases of $\ker(T)$ and $\text{range}(T)$. Verify the rank-nullity theorem.

[4]

5. Does there exist a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $T(1, 1, 1) = (0, 1, 1)$, $T(0, 1, 1) = (0, 0, 1)$, and $T(0, 0, 1) = (1, 1, 1)$? If yes, then find T and the matrix representation of T with respect to the basis $\{(0, 1, 0), (1, 0, 0), (0, 0, 1)\}$ for both domain and codomain.

[6]

6. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. If $A^{-1} = \frac{5}{2}I + kA + \frac{1}{2}A^2$, where I is the 3×3 identity matrix, then find the value of k . Hence or otherwise, find the eigenvalues of A^{-1} .

[4]

7. Let $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$. Then

[4]

- (a) Find the eigenvalues and the corresponding eigenvectors of A .
- (b) Find the algebraic and geometric multiplicities for each eigenvalue of A .